

Task2_AI_ML_Internship_California_Housing_Dataset Content

Methodology

The project follows a professional industry-aligned Machine Learning pipeline:

- **Data Preparation:** Load the California Housing dataset and separate the target variable (Median House Value) from input features.
- **Feature Scaling:** Apply StandardScaler to the input features. This critical step ensures all features have a common scale, preventing some features from dominating others and improving model stability.
- **Model Selection:** Train multiple algorithms to identify the best-performing model:
 - **Linear Regression:** Used as a baseline for performance.
 - **Ridge Regression:** Employed to help reduce overfitting.
 - **Decision Tree:** Utilized to capture non-linear relationships in the data.
- **Evaluation Strategy:** Split data into training and testing sets (typically 20% for testing) to evaluate models on unseen data.

◆ Comparison Table

Model	Purpose	Performance Indicators
Linear Regression	Baseline model	RMSE ↓, R ² ↑
Ridge Regression	Better generalization	RMSE ↓, R ² ↑
Decision Tree	Non-linear relationships	RMSE ↓, R ² ↑

◆ Results

- **Linear Regression:** Provides baseline accuracy.
- **Ridge Regression:** Typically improves generalization, slightly better RMSE/R² than baseline.
- **Decision Tree:** Captures complex patterns, but may overfit if not tuned.
- **Best Model:** The one with lowest RMSE and highest R² on test data.

- **Visual Validation:** Scatter plot of *Actual* vs. *Predicted* shows alignment with red reference line → confirms predictive accuracy.

◆ **Conclusion**

- The pipeline ensures fairness and stability in evaluation.
- **Final model selection** is based on lowest RMSE and highest R^2 , supported by scatter plot validation.
- Demonstrates a professional, industry-aligned workflow for machine learning projects